

High Level Design (HLD)

AI Student Evaluator.5

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Document

Attribute	Details
Document Name	AI Student Evaluator – High Level Design
Version	1.0 (Final Submission)
Author	Utkarsh Yadav
Audience	Competition Judges, Software Engineers, QA Engineers
Backend	Java 21, Maven, Javalin
Frontend	React, Recharts, jsPDF
AI Engine	Google Gemini 2.5 Flash

Revision History

Version	Date	Author	Description
0.1	04 July 2026	Utkarsh Yadav	Final Draft
0.5	04 July 2026	Utkarsh Yadav	Architecture Defined
1.0	04 July 2026	Utkarsh Yadav	Competition Submission

1. Summary

The AI Student Evaluator is an intelligent learning analytics platform that automatically evaluates how well students understood a classroom session. The application compares a mentor's reference summary against students' WhatsApp discussions using Google Gemini 2.5 Flash. Instead of relying on keyword matching, it performs deep semantic evaluation to generate objective scores, identify proven strengths and missing gaps, and produce actionable mentor recommendations. The evaluated data is transformed into an interactive dashboard and exportable PDF reports.

1.1 Business Value & Problem Statement

Mentors spend considerable time manually reviewing classroom discussions to determine comprehension. This manual evaluation is subjective, time-consuming, and difficult to scale across larger cohorts. By automating this process using Large Language Models, this solution improves mentor productivity, standardizes student evaluations, and provides data-driven insights that are otherwise difficult to obtain.

1.2 Objectives

- Automate student evaluation by comparing teaching intent with student understanding.
- Generate structured, highly accurate AI feedback via a strict JSON schema.
- Produce class-level analytics and rank students.
- Help mentors easily identify knowledge gaps and generate next-step learning goals.
- Provide an extensible architecture for future enhancements (e.g., LMS integration).

2. Project Scope

2.1 In Scope (Version 1.0)

- Mentor summary and WhatsApp chat ingestion (Text-based).
- Batch AI evaluation using Gemini 2.5 Flash.
- Student ranking and metric classification.
- Interactive React analytics dashboard.
- Client-side PDF report generation.

2.2 Out of Scope (Future Roadmap)

- Direct document ingestion (PDF/PPT/DOCX).
- LMS integration and Cloud Deployment.
- Historical multi-session trend analysis.
- Authentication and Role-Based Access Control (RBAC).

3. Technology Stack & Rationale

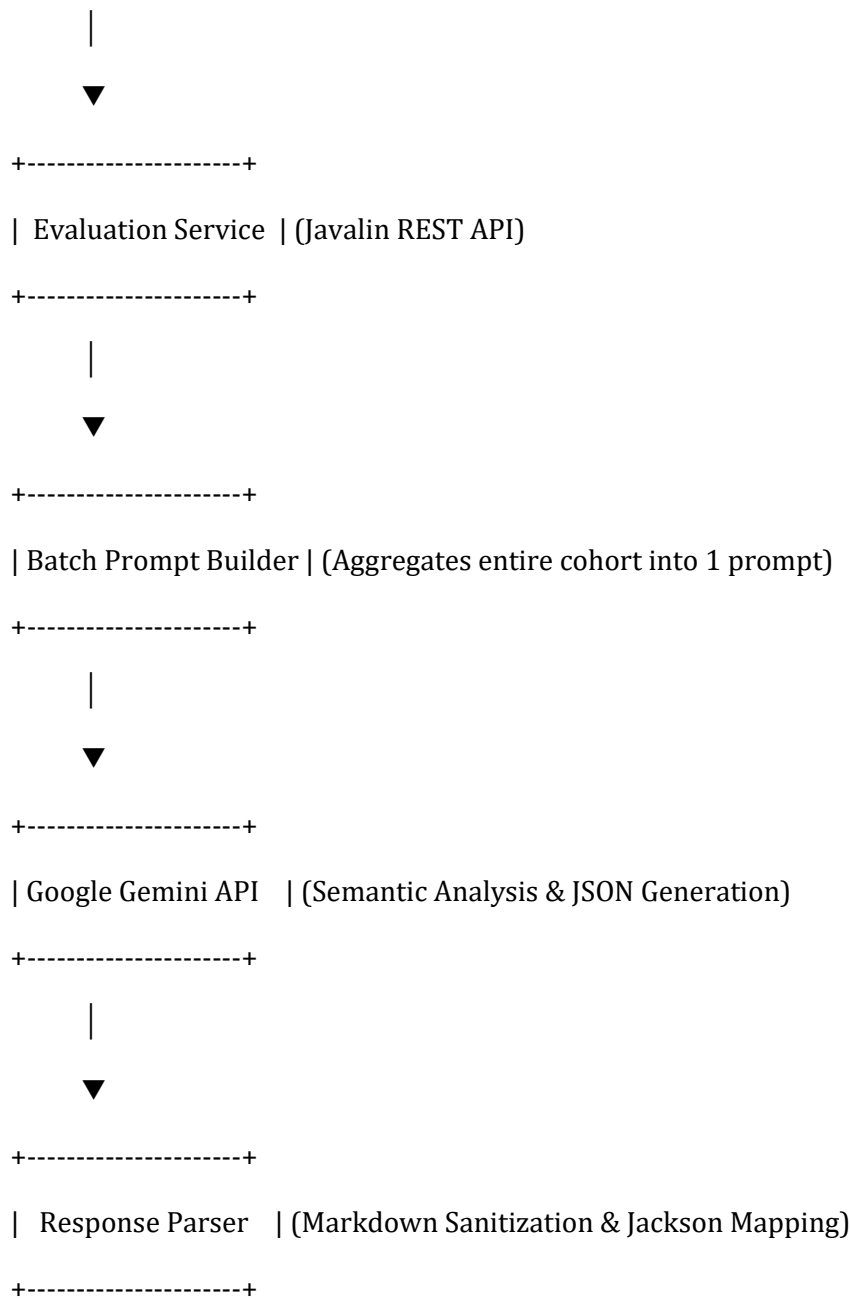
Layer	Technology	Reason
Backend	Java 21 + Maven	Modern LTS, modular architecture and dependency management
AI	Google Gemini 2.5 Flash	Fast semantic evaluation with structured JSON output
Frontend	React	Component-based responsive UI
Charts	Recharts	Interactive analytics visualizations
Icons	React Icons	Consistent UI iconography
JSON	Jackson	Reliable serialization/deserialization
HTTP	Java HttpClient	Native lightweight REST integration
Mentor Summary	Tactiq.ai	Automatic detailed summary generation

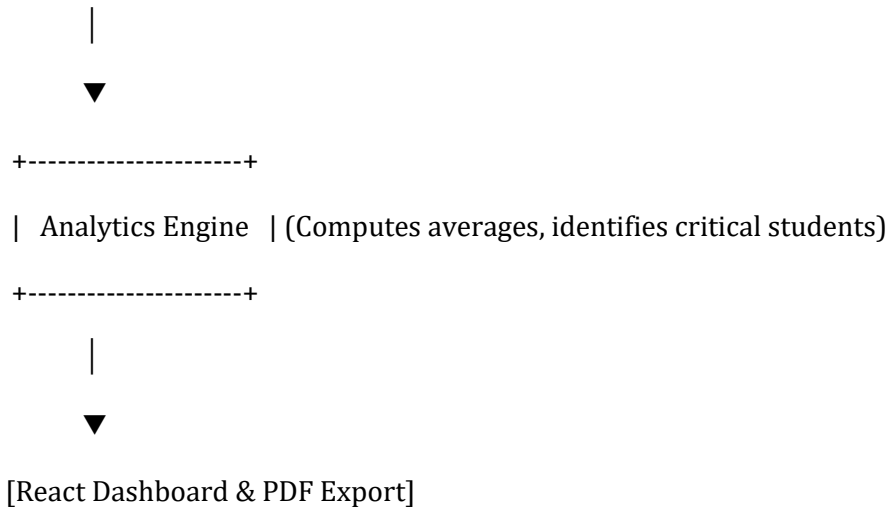
4. System Architecture

The system utilizes a decoupled Client-Server Architecture. The frontend relies purely on a structured JSON contract provided by the backend, ensuring independent scalability.

4.1 High-Level Data Flow

[Mentor Input: Summary (tactiq.ai) & Chat Log]





4.2 Component Responsibilities

Component	Responsibility
Main / API Gateway	Application entry point; exposes /api/evaluate REST endpoint.
EvaluationService	Coordinates the complete batch evaluation workflow.
PromptBuilder	Dynamically constructs a single Master Prompt injecting the session payload and forcing a strict JSON schema output.
AIService	Invokes the external Google Gemini API.
ResponseParser	Aggressively sanitizes AI output and maps JSON to Java Domain Objects.
ClassAnalyticsService	Computes overarching cohort metrics (Averages, Highest/Lowest scores).
React Dashboard	Renders dynamic Recharts and metric matrices based on the JSON contract.
Mentor Summary	Automatically summary generated with the use of tactiq.ai

4.3 Prompt Engineering Strategy

The prompt instructs the LLM to compare the student's understanding only against the mentor summary. The model returns structured JSON containing score, understanding summary, strengths, missing concepts, one-line mentor recommendation, next learning goal and feedback.

The response format is constrained to simplify parsing and improve consistency.

4.4 Student Scoring Methodology

Scores are generated by the LLM based on semantic understanding rather than keyword matching. The evaluation considers concept coverage, technical correctness, completeness and alignment with the mentor summary. The backend treats the AI output as structured evaluation data and uses it to calculate class analytics.

5. System Requirements

4.1 Functional Requirements

ID	Requirement	Priority	Acceptance Criteria
FR-01	Read Inputs	High	Mentor Summary and Chat Log loaded successfully.
FR-02	Batch AI Evaluation	High	Cohort is evaluated in a single API request, bypassing rate limits.
FR-03	Dashboard Generation	High	DashboardData payload successfully powers the React UI.
FR-04	PDF Export	High	Professional, multi-page PDF generated locally without data loss.

4.2 Non-Functional Requirements

Category	Requirement	Target
Performance	Efficient execution	Cohort evaluation completes in < 15 seconds.
Reliability	Fault tolerance	Aggressive regex string cleaning prevents JSON parsing crashes.
Maintainability	JSON Contract	Frontend is strictly decoupled from backend logic.
Security	API Protection	Environment variables utilized for Gemini API keys.

6. Domain Model & JSON Contract

The backend exposes a stable DashboardData JSON object. This separation of concerns allows the React UI to evolve entirely independently of the Java evaluation pipeline.

Core Entities

- **StudentEvaluation:** studentName, score, confidence, understandingSummary, strengths (List), missingConcepts (List), mentorRecommendations, nextObjective.
- **ClassAnalytics:** totalStudents, averageScore, highestScorer.
- **DashboardData:** Wraps ClassAnalytics and List<StudentEvaluation>.

6.1 Backend Package Structure

```
com.ai.mentor
├── ai
├── config
├── export
├── model
├── parser
├── service
├── util
└── resources
```

6.2 Module Design

Module	Description
AI Module	Responsible for communicating with Google Gemini, sending prompts and returning structured JSON.
Parser Module	Parses WhatsApp chat into StudentChat objects and converts AI JSON into StudentEvaluation objects.
Service Layer	Coordinates business logic including evaluation, analytics and dashboard generation.

Model Layer	Represents domain entities such as StudentEvaluation, StudentChat, DashboardData and ClassAnalytics.
Export Layer	Exports dashboard data and reports for external consumption.
Configuration	Stores application constants and runtime configuration.

7. Architecture Decision Record (ADR) & Trade-offs

Decision	Alternative Considered	Reason Chosen	Trade-off / Benefit
Batch Processing Evaluation	Sequential Evaluation (Looping)	Free-tier LLM APIs have strict per-minute/per-day limits.	Benefit: Prevents 429 Resource Exhausted crashes and executes significantly faster by processing the entire class in one network call.
Gemini 2.5 Flash API	Local LLM (Ollama)	Superior semantic reasoning and massive 1M token context window.	Trade-off: Requires internet connectivity, but guarantees highly accurate, structured output.
Decoupled React SPA	JavaFX / Thymeleaf	Provides a superior, modern web experience and interactive SVG charting.	Benefit: Highly reusable components; easily adaptable for future cloud deployment.
Text-Based Ingestion	PDF/DOCX Uploads	Simplifies the MVP scope to focus heavily on AI accuracy.	Trade-off: Requires users to copy-paste inputs for version 1.0.

8. Dashboard Architecture

The dashboard consumes the DashboardData JSON produced by the Java backend. The frontend is completely decoupled from the evaluation engine. Every UI component is driven by structured JSON, allowing the backend and frontend to evolve independently.

Dashboard Component	Purpose
Summary Cards	Display total students, average score, highest and lowest score
Score Distribution	Visualize score ranges using charts
Performance Table	Rank students with search and sorting
Student Detail Panel	Display strengths, gaps, recommendations and feedback
Needs Attention	Highlight students requiring mentor intervention
PDF Export	Generate a downloadable evaluation report

8.1 User Experience (UX) Principles

Present the most important insights first (summary cards).
Enable rapid identification of students requiring attention.
Provide detailed drill-down only when required.
Maintain a clean and responsive interface.
Minimize clicks to reach actionable information.
Support printable reports through PDF export.

8.1.1 Assumptions

- Mentor summary accurately reflects the session.
- Students contribute meaningful WhatsApp messages.
- Gemini responses follow the requested JSON schema.
- Internet connectivity is available during evaluation.

9. Innovation Highlights

9.1 Benefits of Current Architecture

The chosen architecture allows independent enhancement of the AI provider, parser, dashboard and analytics modules. Future additions such as multi-provider AI support, additional input formats and cloud deployment can be introduced with minimal impact on existing components.

This solution goes significantly beyond traditional keyword-based assessment tools by leveraging generative semantic AI. Key innovations include:

1. **Contextual Grading:** AI-generated conceptual scoring instead of rigid keyword matching.
2. **Rate-Limit Immunity:** A customized Batch Processing architecture that protects the application from API quota exhaustion.
3. **Actionable Intelligence:** Automatic identification of knowledge gaps and personalized next-step learning goals.
4. **Zero-Latency Export:** Client-side rendering of multi-page PDF reports, saving backend bandwidth and processing power.

10. Limitations & Future Enhancements

10.1 Current Limitations

- Supports text-based mentor summaries only.
- Accepts pasted WhatsApp chats rather than direct imports.
- Evaluates one session at a time.
- AI response quality depends on the provided mentor summary.
- Relies on tactiq.ai for Mentor Summary which is billable.

10.2 Readiness for Future Enhancements

The modular architecture allows future integration of:

- Multiple AI providers (Gemini, OpenAI, Ollama)
- Additional document formats (PDF, PPT, DOCX)
- Historical trend analysis
- LMS integration
- Authentication and role-based access
- Cloud deployment